

POKHARA UNIVERSITY
FACULTY OF SCIENCE AND
TECHNOLOGY
SCHOOL OF ENGINEERING

Exam	Final Internal Examination 2026 Fall		
Level	B.E.	FM	100
Program	Computer	PM	45
Year/ Part	IV/II	Time	3 Hrs

Subject: Cloud Computing and Virtualization

*Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicates full marks. Assume suitable data if necessary.
Attempt all the questions.*

- 1 a) Define cloud computing according to NIST. Explain the need for cloud computing with suitable real-world applications. 7
- b) What is Cloud Reference Model and describe the NIST Cloud Architecture with a neat diagram. 8
- 2 a) Describe the cloud service models with suitable examples. 8
- b) Explain the design principles of cloud architecture and discuss the Jericho Cloud Cube Model. 7
- 3 a) Explain virtualization technologies. Compare Type-1 and Type-2 hypervisors. 8
- b) What is containerization and give an overview of Kubernetes architecture and components. 7
- 4 a) Explain VPC and VPN. How are they used to secure cloud networks? 7
- b) Explain scaling strategies in cloud computing. Differentiate horizontal and vertical scaling.

OR

What is Autoscaling? Provide the steps for autoscaling. Take reference of any CSP. 8

- 5 a) Describe cloud storage models. Explain object, file, and block storage with examples. 7
- b) Explain Database as a Service (DBaaS). Compare SQL and NoSQL databases used in cloud computing. 8
- 6 a) What is the cloud threat? Explain with one real cloud data breach example.

OR

Describe cloud security dimensions. 7

- b) Explain the concept of Hashing in the cloud. How does it work? 8

7 Write Short notes. (Any two) 2X5

a) SLA

b) Multihoming

c) Infrastructural Constraints in Cloud